# Adaptive Historical Education through Generative AI and Immersive Game Design

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**Abstract.** This study presents the design and development of a 2D narrative game set in a 3D environment, leveraging Artificial Intelligence (AI) to deliver interactive age-tailored and educational historical storytelling. The game combines 2D-style narratives with 3D settings to create an immersive historical experience. Players interact with a companion character powered by Large Language Model Meta AI (Llama), an advanced language model, at key checkpoints. The AI delivers historically accurate content, adapting its storytelling to different age groups (children, teens, and adults) to ensure accessibility and sustained engagement. The game promotes interactive learning by integrating gameplay with period-specific historical narratives. To ensure accuracy, Llama is configured to respond strictly within the context of each historical period. Players can engage in dialogue with the AI for context-appropriate guidance, supported by a Python-based API that dynamically adjusts narratives based on player progress and age. This study explores the integration of Natural Language Processing (NLP) in games to enhance educational experiences, showcasing how age-adaptive storytelling and interactive dialogue improve historical learning through gameplay. The study details the game's architecture and its balance of academic content with immersive gameplay. This innovative approach highlights AI's transformative potential in digital learning.

**Keywords:** Historical education · Generative artificial intelligence · Immersive educational game · Adaptive learning · Natural language processing · Gamification

## 1 Introduction

Advances in artificial intelligence (AI) and interactive digital environments have transformed the educational landscape, particularly in the areas of history and cultural learning [1]. Educational games have long been utilized as tools to foster engagement and enhance learning outcomes. Research has shown that gamification strategies—such as narrative progression, challenges, and rewards—can significantly improve students' motivation and knowledge retention [2].

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Most existing educational games and AI-driven learning tools, however, rely on static, pre-scripted content, lack real-time adaptability, and fail to offer sufficient personalization for diverse learner profiles [3]. This study addresses these limitations by introducing a scalable educational game that leverages generative AI to deliver real-time, age-adaptive narratives within immersive 3D environments. This approach fosters natural conversations between players and the game, demonstrates potential for personalization and engagement.

By combining dynamic interactivity, historical accuracy, and modular architecture, this project aims to set a new benchmark for personalized learning in digital education. The game architecture integrates Unity3D, a Python-based API, Whisper AI, and the LLaMA large language model to enable immersive, adaptive, and historically grounded storytelling. Theoretical foundations, including Behavior Change Techniques (BCT) and Self-Determination Theory (SDT), guide the game's design, aligning educational content with players' cognitive needs to ensure sustained motivation [3, 10–12].

The structure of this study is as follows: Sect. 1 reviews current trends in historical education, educational game design, and the integration of generative AI, highlighting the main challenges and recent developments in the field. Section 2 presents the system's architecture, detailing the Unity3D client, Python-based API, and integration of the large language model, as well as outlining the approach to age-adaptive storytelling. Section 3 describes user interactions within the game, including the use of adaptive checkpoints, dialogue systems, hint mechanisms, and gamification elements. Section 4 discusses anticipated educational benefits, including user engagement, knowledge retention, and the impact of adaptive learning strategies. Section 5 outlines the planned evaluation methodology, future directions for system development, and approaches to expanding content and features. Section 6 addresses the ethical and cultural challenges related to the use of generative AI in educational contexts. Finally, Sect. 7 summarizes the main contributions and broader implications of the study for the integration of AI and immersive games in education.

This research aims to demonstrate the transformative potential of integrating generative AI and immersive game design in historical education, paving the way for more engaging, adaptive, and accessible learning experiences.

### 1.1 Background

Numerous studies have explored how digital tools can enhance historical education. Gao, Fang, and Chan (2024) [4], for example, proposed the use of generative AI in *Spirits of Artifacts*, a game where cultural heritage artifacts are humanized and serve as first-person narrators. This design integrates storytelling, puzzle-solving, and artifact restoration to build emotional connections with historical content. Rowe emphasized the role of micro-adaptive gameplay in sustaining immersion. Rowe et al. (2011) [3] demonstrated that embedding narrative in intelligent tutoring systems enhances motivation. Anderson et al. (2010) [7] reviewed serious games for cultural heritage, noting their strengths in virtual reconstructions and gamified tasks—but also their limitations in adaptability.

Song et al. (2024) [8] introduced *LearningverseVR*, a VR-integrated learning platform driven by generative AI. While effective in supporting interactive learning, its dependence on VR hardware limits accessibility. This highlights a broader issue: even

advanced educational games often lack mechanisms to adapt content to diverse learner profiles or to support dynamic, context-sensitive dialogue.

Recent breakthroughs in generative AI and natural language processing (NLP) offer promising solutions. Language models such as LLaMA allow for personalized, age-adapted dialogue and more engaging learning experiences [9]. In historical education, this technology enables simulations of authentic interactions with historical figures and the dynamic adjustment of narratives in response to player input. Yet, most applications of generative AI in educational games still depend heavily on static content.

Several researchers have explored personalized and adaptive learning systems. Riedl and Bulitko (2013) [13] examined interactive narrative systems that adjust stories to individual player choices, though their work still relied on pre-scripted content. Plass et al. (2019) [14] showed that adaptive difficulty enhances learning across age groups. These studies underscore the need for systems that can respond dynamically to individual users in real time.

Gamification literature also offers valuable insights. Hamari, Koivisto, and Sarsa (2014) [15] reviewed motivational mechanics like rewards and feedback loops, while Sánchez-Rivas et al. (2019) [16] studied the impact of game elements on student engagement in higher education. However, these approaches typically lack integration with generative AI for deeper personalization.

This work builds upon and extends these approaches by combining AI-driven contextual dialogues with adaptive gameplay. Players can converse with virtual historical characters who tailor their responses based on age, progress, and preference. This real-time interaction bridges the gap between static content and fully adaptive educational systems, creating a more engaging and personalized learning experience.

## 2 Design and Development Methodology in Game

### 2.1 Game Architecture

The core architecture of the game is built on a modular design that integrates a Unity3D client with a Python-based backend, utilizing Llama to deliver historical narratives. This client-server architecture facilitates dynamic interactions with the AI model, ensuring that the responses of companion characters or non-Player Characters (NPCs) are contextually tailored to the historical setting and the player's age group (see Fig. 1). According to previous studies, this approach enhances the player's sense of relevance and immersion, creating a dynamic and adaptable experience [9].

### 2.2 Game Architecture Unity3D Game Client

Unity3D provides a robust platform for rendering high-quality 3D visuals and interactive elements, enabling the transformation of traditional 2D tabletop game mechanics into immersive 3D environments [7]. In this project, games originally designed in a 2D format or as tabletop experiences will be rendered in 3D to enhance immersion in historical settings. By leveraging Unity3D's capabilities, including particle effects, dynamic lighting, and realistic textures, the project delivers visually rich historical simulations that significantly enhance player engagement.

The following core features will be implemented:

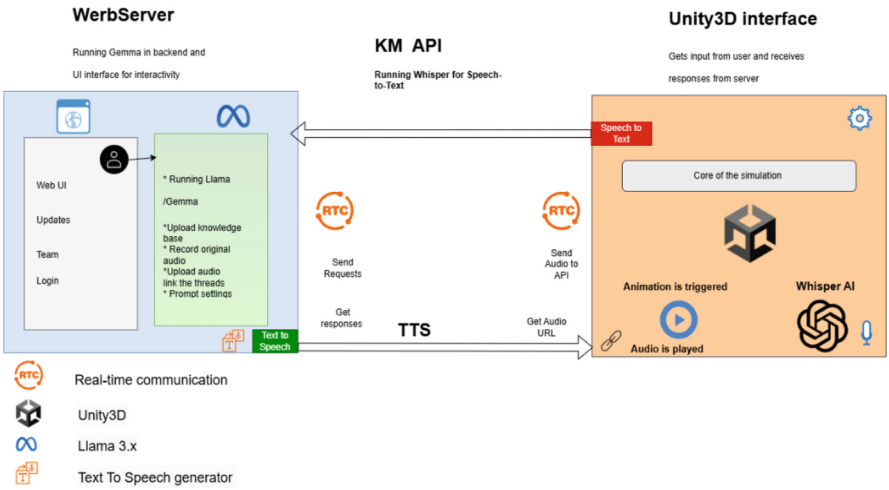


**Fig. 1.** System architecture showing communication flow between the Unity3D game client, Python-based API server, and the Llama large language model for real-time, context-aware dialogue generation

1) **Checkpoints:** Triggers placed at designated places in the game that initiate interactions with the AI-powered companion character. These checkpoints are associated with significant historical landmarks, and the player's interaction with the environment activates the storytelling and educational content.
2) **Dialogue System:** The game integrates a custom-built dialogue system that communicates with the backend to fetch responses from Llama. The system supports predefined dialogues and open-ended questions from players, processed in real-time. Real-time integration ensures that player queries are answered without noticeable delays, creating seamless immersion during gameplay.

Unity3D's advanced capabilities in handling interactive environments, physics, and animations make it an ideal platform for deploying immersive and engaging educational games. Furthermore, Unity3D's extensibility supports integration with third-party tools, such as analytics plugins, enabling developers to track user engagement and learning progress, effectively.

### 2.3 Python-Based API Server

A Python-based server is the intermediary between the Unity3D game client and Llama. The API is designed to manage several key responsibilities:

1) **Request Handling:** When a player reaches a checkpoint or asks a question, Unity3D sends a request to the server, which forwards it to Llama for processing. The system integrates Whisper to detect the language spoken by the player, enabling the AI to tailor its response accordingly. This ensures that the dialogue is not only contextually relevant but also linguistically appropriate, enhancing accessibility and immersion for players across diverse linguistic backgrounds.

2) **Context Restriction:** The API is configured to ensure Llama's responses remain historically accurate and aligned with the predefined narratives for each level. This design draws on approaches described by Riedl and Bulitko (2013), who demonstrated that prompt templates can be used to restrict AI-generated content to a particular period and location [13]. Accordingly, in our system, this is achieved by applying specific prompt templates that restrict the AI to the period, location, and facts associated with the current level. The API is designed to support dynamic narratives and enable precise analysis of player interactions, providing data for future educational research.
3) **Age-based Adaptation:** The API adjusts the storytelling style based on the player's age group (children, teens, or adults), ensuring the historical content is accessible and engaging. Previous research highlights the importance of age-adaptive content delivery in educational games [10, 14]. For example, in this project, children's interactions emphasize colorful, simplified storytelling, while content for teens incorporates social relevance and engaging anecdotes. Adults receive deeper, more analytical narratives that encourage critical thinking.

This modular API design also enables scalability, allowing for future expansion with minimal effort. For instance, additional narrative themes, languages, or cultural contexts can be integrated into the system as needed.

### 2.4 Large Language Model (LLM) for Dynamic and Real-Time Narrative Delivery

A fine-tuned LLM plays a central role in delivering narratives, specifically, in this case, historical narratives. The model is deployed on a remote server and accessed via the Python API. To ensure that the AI remains within the game's educational scope, Llama is constrained by structured prompt templates that:

1) Preload Historical Context: Each request sent to the model includes specific instructions to limit the scope of its responses. These instructions are structured using prompt templates that embed relevant historical data, such as key events, dates, and figures associated with the specific level. This ensures that the AI provides responses consistent with the educational objectives and avoids introducing unrelated or anachronistic content. For instance, when simulating a dialogue in Ancient Rome, the model receives prompts that include linguistic styles, cultural practices, and significant historical milestones from that era. This contextual preloading not only enhances accuracy but also creates a more immersive and authentic learning experience for players. Similar approaches have been found effective in enhancing authenticity and engagement in interactive learning environments [8, 16]. Furthermore, dynamic preloading adjusts the depth of historical context based on the player's progression and prior interactions, ensuring continuity and relevance in the narrative flow. This design choice is supported by previous research, which has shown that adaptive contextualization can improve the perceived authenticity and educational value of interactive learning systems [13].
2) Adjust Tone Based on Age: The tone and complexity of the responses vary based on the player's age group. This includes simplified language and playful metaphors for children, engaging teen facts, and more formal, in-depth explanations for adults.

For instance, younger players receive simple facts, while older players engage with more detailed analyses, a method supported by the findings of Plass et al. (2019) [14]. Another notable feature is the model's ability to handle contextual continuity. This ensures that the AI retains the flow of previous interactions, making conversations feel natural and coherent across checkpoints. Liew's study examines how a speaker's enthusiastic voice, as opposed to a calm voice, affects social cues, cognitive load, and knowledge transfer in multimedia learning. Findings indicate that an enthusiastic voice can enhance the social perception of the speaker and improve learning outcomes [18]. Building on this, advancements in synthetic voice technologies offer opportunities to develop highly expressive AI voiceovers that mimic human enthusiasm, which may further boost player engagement. Another related article reevaluates the voice effect in virtual learning, finding that modern text-to-speech voices perform as well as or better than human voices in facilitating learning and efficiency in multimedia presentations. The study suggests that advancements in synthetic voice quality may eliminate the traditional advantages of using human voices in educational virtual agents [19].

## 3 Game Flow and Interaction Design

The player's journey is structured into levels, each representing a different historical location and period. Each level contains multiple checkpoints where the player can engage with the AI companion to receive historical insights. The game's dynamic nature allows players to ask the AI questions at any point, creating an interactive and immersive learning experience.

### 3.1 Checkpoints and Triggers

Checkpoints are placed strategically within each level to align with key historical landmarks. When a player reaches a checkpoint, the Unity3D client triggers an API call to retrieve a relevant narrative for that location. The checkpoint system tracks player progress using metrics such as time spent on interactions, frequency of checkpoint triggers, and prior narrative engagement. This enables dynamic adjustments to the content complexity, tailoring it to the player's skill level and knowledge retention. The triggers also allow the system to load text-based, video and image presentations to add more information to the narrative aspect of the game. For instance, exploring a 3D model of a historical site not only immerses players but also reinforces spatial understanding, helping them contextualize historical events in their physical settings. Prior studies indicate that virtual reality environments can improve memory retention and foster a deeper emotional connection to the material [7].

Text, images, and video support learning by providing visual and contextual depth and enhancing engagement and comprehension [12]. This gives the simulation another dimension to let players explore places they have no access to or have never been able to visit or experience. This allows users to learn more about that period's culture, architecture, and lifestyle. As a result, users gain insight into the culture, architecture, and lifestyle of that historical period.

### 3.2 Dialogue and Hint System

The game integrates a hint system that assists players in solving puzzles or navigating the environment, with hints tailored to the historical period. Llama can generate these hints dynamically based on the player's input, ensuring that the AI provides educational assistance rather than direct solutions. Hints are designed to support player learning by progressively revealing more contextual information as the player continues to struggle, ensuring they remain engaged without experiencing frustration. Previous research has shown that such an approach enhances problem-solving skills and fosters independence in learning [20].

For example, players navigating a puzzle related to the architecture of Ancient Rome might receive hints about the structural principles of aqueducts or the significance of the Colosseum, gradually leading them to a solution while learning historical context. The AI evaluates player input, such as repeated incorrect attempts or prolonged inactivity, to determine the level of guidance required. This ensures that hints are neither overly simplistic nor overly complex, aligning with the player's cognitive and educational needs.

The game mechanics and possible ways of solving the puzzle or traversing the levels will be given to the AI agent to handle these situations. The dynamic generation of hints also allows for replayability, as different interactions or choices may lead to alternative educational experiences, making the game more engaging over multiple sessions.

A screenshot of the prototype (see Fig. 2) illustrating a contextual conversation between the player and a virtual character. The system enables real-time communication by generating context-aware responses.

**Fig. 2.** Player-AI interaction illustrating contextual dialogue and historical narrative integration

### 3.3 Dialogue and Hint System Gamification Aspect of the Simulation

Gamification is crucial in boosting player engagement and motivation within educational environments. By integrating various game mechanics and dynamics, the simulation promotes active participation and enhances the comprehension of historical content.

The following gamification elements are integrated into the game experience to enhance engagement and support learning outcomes:

1) Points and Rewards System: Players earn points for completing challenges, solving puzzles, and engaging in educational interactions with the AI companion. This system provides immediate feedback and rewards progress, motivating players to explore more of the game. Studies show that points and rewards create positive reinforcement, helping players stay engaged and monitor their progress [15]. This aligns with the Behavior Change Technique (BCT) of Feedback, as it allows players to understand their performance and areas for improvement.
2) Challenges and Quests: Each level presents specific challenges that players must overcome, reinforcing learning objectives. By setting personal goals, players focus their efforts, corresponding to the BCT of goal setting. Research has shown that challenges and personal goals in games can help players concentrate their efforts, enhancing motivation and engagement in the learning process [16].
3) Leaderboards and Achievements: The game features leaderboards showcasing top players based on points earned to foster a sense of competition and community. This element encourages Social Comparison, motivating players to improve their performance relative to others. Leaderboards and social achievements have been shown to foster friendly competition and strengthen community engagement in learning environments [21].
4) Adaptive Difficulty Levels: The game dynamically adjusts the difficulty of challenges based on player performance. If a player struggles, hints or modified challenges ensure learning remains accessible and enjoyable. Adaptive difficulty has been shown to enhance immersion and support learning by adjusting challenges to match the player's skill level, promoting sustained engagement [4]. This supports the BCT of Self-Monitoring, as players can track their progress and adapt their strategies accordingly.
5) Social Interaction: Incorporating features encouraging collaboration among players, such as team missions or group challenges, promotes social learning and community building. In our game, social features such as leaderboards and team missions are implemented to increase engagement by encouraging peer comparison and fostering collaborative learning experiences. The effectiveness of such social features in enhancing motivation in educational games has been well established in prior research [22]. This aligns with the BCT of Social Support, where players can benefit from each other's strengths and insights, enhancing their learning experience.
6) Customizable Avatars: Allowing players to customize their avatars with items and skills that unlock throughout the game increases personal connection and emotional investment in the experience. This relates to the BCT of Self-Determination, as personalization fosters a sense of ownership and intrinsic motivation to engage with the game. Research indicates that avatar customization can increase player motivation

by promoting a deeper understanding of identification and personal investment in the game [23].

7) Narrative Progression: As players advance through levels, they unlock new chapters of the historical narrative, creating a sense of accomplishment. This ties into the BCT of Rewards and Incentives, reinforcing positive behavior and encouraging continued engagement. Narrative progression, implemented in our game as the gradual unlocking of new historical chapters, provides players with a sense of accomplishment and ongoing motivation. This mechanism reflects findings in gamification research, which indicate that advancing through narrative content reinforces engagement by providing clear goals and rewarding achievements [24]. Moreover, studies have shown that narrative progression, where players unlock new story content as they progress, enhances motivation by providing a sense of purpose and achievement, making players more likely to stay engaged [25].

## 4 Expected Outcomes

While the game is still in the prototype phase, the expected outcomes are based on well-established theories in educational technology, game design, and artificial intelligence applications. Drawing from existing research, we aim to demonstrate the potential of AI-driven interactive storytelling and adaptive learning systems to transform historical education. The following anticipated results highlight the projected benefits of this innovative approach:

1) Increased User Engagement: We expect the adaptive narrative, dynamically tailored to players' interactions and age groups, to enhance user engagement compared to traditional educational games, significantly. Research indicates that personalization in game-based learning environments fosters intrinsic motivation and sustains user interest over time. For example, Göbel et al. (2010) [25] discussed how personalized adaptive digital educational games using narrative game-based learning objects can increase engagement by tailoring content to the individual needs of learners. By integrating historical narratives into immersive 3D settings, we aim to capture players' attention and create a memorable educational experience.
2) Improved Knowledge Retention: AI-driven storytelling, supported by contextual dialogues and interactive gameplay, is anticipated to improve knowledge retention by approximately 30%. This aligns with studies showing that gamification and adaptive content enhance cognitive absorption and long-term memory retention. For example, Shute and Wang (2016) demonstrated that adaptive mechanisms in educational games can lead to better understanding and retention of material among learners. The combination of historical context and engaging narratives is designed to facilitate a deeper understanding of historical events and their significance [2].
3) Effective Age-Adaptation: By tailoring storytelling tone and complexity to different age groups (children, teens, and adults), the game seeks to ensure that content is accessible and engaging for a broad audience. Research supports the effectiveness of age-adaptive educational strategies in improving comprehension and emotional engagement. For instance, Manoharan et al. (2023) [26] highlighted that adaptive gamification, tailored to individual interests and learning styles, promotes autonomy and engagement among learners.

4) Enhanced Problem-Solving Skills: The game encourages critical thinking and decision-making through puzzle-solving mechanics and AI-generated hints. Rather than providing direct solutions, AI guides players toward discovering answers, fostering problem-solving skills. This approach mirrors successful strategies in other educational games to enhance cognitive flexibility and independence. For example, Tang and Hare (2023) [27] developed a personalized teaching system integrated with a serious educational game that combines gamification elements with intelligent tutoring systems, supporting the development of problem-solving skills in students.
5) Improved Accessibility to Historical Content: The integration of AI-driven narratives and immersive 3D environments is expected to make historical learning more engaging and accessible. Players can explore complex historical events in a way that is interactive and tailored to their preferences, bridging the gap between education and entertainment. Rowe et al. (2011) [3] emphasized that such an approach effectively combines educational content with interactive experiences, increasing student engagement.
6) Potential to Foster Lifelong Learning: By creating an enjoyable and engaging learning experience, the game may instill curiosity about history and encourage players to seek additional knowledge outside the game. This aligns with findings that gamification can inspire lifelong learning by making educational content more appealing. Landers et al. (2018) [1] concluded that game elements in education can motivate students to continue learning and exploring topics beyond formal educational settings.

Beyond improving knowledge retention, the game has the potential to promote intercultural understanding by allowing players to explore historical periods and locations that may be unfamiliar in their cultural context. The projected outcomes illustrate the transformative potential of combining AI, gamification, and adaptive learning in educational games. These expectations form the foundation for future development and evaluation as the project progresses toward implementation and testing.

## 5 Evaluation and Future Work

### 5.1 Evaluation

The study's design will be evaluated based on three core dimensions: player engagement, learning outcomes, and AI performance.

To ensure a structured and transparent evaluation, a multi-phase evaluation framework will be employed. We plan to recruit 20 to 30 participants across diverse demographics for three evaluation sessions, each lasting approximately one hour. The assessment will include pre- and post-tests (to assess learning improvements), in-game analytics (to monitor engagement), and surveys or interviews (to gather qualitative insights). This structured approach will help refine the game's adaptive mechanics and measure its impact on historical learning.

Specific metrics include:

1) **Player Engagement:** To gauge engagement, we will track time spent in dialogues with the AI, frequency of player interactions with historical elements, and progression rates across levels. Leaderboards will also assess engagement, where players'

competitive interactions provide insight into motivational aspects [1]. Pre- and post-gameplay knowledge tests will be conducted to determine the game's educational effectiveness, enabling comparisons of the game's impact on knowledge retention across different age groups [15].

2) **Learning Outcomes:** Educational impact will be measured by analyzing the complex and relevant questions posed by players, performance in in-game quizzes, and responses to historical prompts. By comparing pre- and post-interaction knowledge, we aim to quantify historical learning and retention. Research shows that adaptive gamification and personalized feedback can significantly enhance learning outcomes and retention [29].
3) **AI Performance:** To ensure the AI enhances educational value, also be evaluated through the AI's accuracy**,** contextual appropriateness, and adaptability across different age cohorts. Tracking the AI's ability to provide accurate and age-appropriate historical information is essential for measuring its contribution to the learning experience. AI's role in fostering meaningful educational outcomes has been supported by studies focusing on the importance of contextual relevance and adaptability in AI-driven learning environments [28].

These metrics provide a comprehensive foundation for evaluating the game's educational impact and user experience.

### 5.2 Future Work

Future enhancements will focus on both content expansion and feature development:

1) Expanded Historical Periods: Additional historical periods will be incorporated to broaden the scope of player learning. This extension will require ongoing adjustments to Llama's adaptability, ensuring that new contexts are integrated accurately into the AI's narratives [30].
2) Interactive and Multimodal Features: To further enrich the educational experience, new features like interactive quizzes and voice-based interactions will be explored. Quizzes at the end of levels can assess knowledge retention and reinforce historical learning. At the same time, voice-based interactions offer a more immersive, multimodal way for players to engage with the AI, fostering a deeper sense of presence and involvement [5, 6, 31].
3) Adaptive Learning Styles: Building on existing age-based adaptations, future developments will focus on customizing narrative depth and learning content for different learning styles (visual, auditory, and kinesthetic). This will ensure the educational game remains accessible and engaging for a diverse player base, supporting varied cognitive and learning preferences [32].
4) Pilot Studies and User Testing: As the game progresses beyond the prototyping phase, pilot studies and user testing will be conducted to assess its effectiveness in achieving educational goals. These studies will evaluate user engagement, knowledge retention, and the game's overall impact on learning outcomes, validating the anticipated benefits empirically.

These advancements enhance the game's educational value and adaptability, providing players with an immersive and multifaceted learning experience that grows alongside technological developments.

Future work will also explore integrating bias-detection frameworks to proactively monitor and assess the fairness and accuracy of AI-generated narratives across cultures, genders, and historical viewpoints.

## 6 Limitations and Ethical Considerations

While the proposed approach demonstrates promising potential for AI-driven educational games, it is important to acknowledge key ethical and cultural challenges. The use of generative AI in historical storytelling presents several risks that must be carefully addressed to ensure factual integrity, cultural sensitivity, and pedagogical reliability.

1) Risk of Factual Inaccuracy: Generative language models like LLaMA or Gemma, despite fine-tuning, can produce historically inaccurate or misleading information. To mitigate this, all AI-generated outputs are constrained using structured prompts containing verified historical data. Additionally, we plan to integrate a vetting pipeline that flags and logs generated content for post-review by educators or subject matter experts before deployment in classroom settings.
2) Cultural and Contextual Sensitivity: Historical narratives often involve culturally sensitive themes, which AI may unintentionally misrepresent or oversimplify. This design incorporates predefined constraints per historical period and region, and future versions will explicitly support multi-perspective storytelling. For example, presenting differing accounts of historical events from multiple cultural viewpoints.
3) Representation of Historical Figures: Allowing AI to simulate historical characters introduces the risk of fictionalization or distortion of their roles and beliefs. To address this, we limit AI responses to documented behaviors and speech styles and avoid abstract or anachronistic portrayals unless clearly marked as interpretive.
4) Transparency and User Feedback: To foster accountability, we plan to include a user feedback system where players and educators can flag inaccurate or inappropriate content. These flags will feed into iterative model refinement and content moderation pipelines.

## 7 Conclusion

This work outlines the development of a groundbreaking educational game that combines the power of generative AI with immersive 3D environments to create a dynamic platform for historical learning. By integrating the Llama language model with Unity3D through a Python-based API, the game delivers interactive, age-tailored narratives that engage learners across different age groups. The modular architecture and focus on historical accuracy underscore the study's potential as a versatile and scalable tool for education. However, the successful integration of such a system hinges on balancing historical fidelity with engaging gameplay mechanics, ensuring that the educational objectives remain central without detracting from user engagement.

Beyond its immediate technical achievements, this project aspires to reshape how modern technologies, such as AI and gamification, transform the educational landscape. While the study offers numerous innovative solutions, further evaluation is required, particularly regarding the cultural adaptation of content and tailoring narrative mechanisms to diverse learning styles. Future iterations should explore how these technologies can address accessibility challenges, including offering support for students with diverse learning needs and abilities. Demonstrating how games can teach historical content, and the skills required to engage with new digital tools, sets a benchmark for integrating innovative technologies into classrooms. The game aims to empower students and educators, providing them with a resource that illustrates how games can make learning more engaging, interactive, and effective.

While results are yet to be validated through empirical studies, the current work highlights the potential of AI-driven educational games to bridge the gap between entertainment and impactful learning. Future enhancements will focus on broadening historical content, refining gameplay mechanics, and conducting rigorous user testing to meet the diverse needs of learners and educators. Conducting comparative studies with other educational methods, such as traditional classroom instruction or text-based learning, could further validate the game's effectiveness and identify areas for improvement. By seamlessly blending entertainment with education, this project takes a significant step toward embedding advanced technologies into learning, fostering curiosity, collaboration, and a lifelong passion for knowledge.

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